SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	We got no idea
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 2
Project / Solution Name	AgriGuard
Name (Leader)	Tang Yun Tian
Email & Phone (Leader)	yuntian1003@gmail.com 0124098320
University Affiliation(s)	Universiti Sains Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro and small agricultural businesses in Malaysia, such as smallholder paddy farms, durian orchards, and commercial nurseries, rely heavily on general workers for daily crop scouting. However, field workers rarely possess the pathology training needed to distinguish Bacterial Leaf Blight from nutrient deficiencies, leading them to send blurry photos via WhatsApp to the owner. The owner typically inspects the field the next day, diagnoses the condition, discovers the required treatment is out of stock in the shed, and waits another 24 hours for agricultural chemical suppliers to reply. During this 48–72 hour lag, pathogens spread aggressively.

This delay directly threatens regional food security and smallholders' livelihoods. Historical outbreaks of rice blast in Malaysia have impacted over 40% of planted hectares across Perak, Penang, Selangor, and Kedah, causing estimated yield losses between 10% and 50%. Climate change is intensifying unseasonal rainfall and pest cycles, while persistent rural labour shortages mean farm owners cannot be physically present in every plot.

Existing software does not address this operational reality: generic spreadsheets cannot identify diseases, enterprise precision-agriculture platforms cost thousands of Ringgit and overwhelm small teams, and consumer chatbots cannot check physical warehouse stock or enforce purchase controls. Slow treatment severely erodes margins for small-scale food producers (SDG 2.3) and undermines agricultural resilience (SDG 2.4).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AgriGuard is an end-to-end, AI-assisted crop disease triage and supply procurement web application tailored for ASEAN smallholder farms. It bridges the critical 48-hour gap between field anomaly detection and chemical treatment delivery through a structured three-role workflow:

Worker: Captures an in-field photo of a symptomatic leaf. An on-device TensorFlow.js MobileNet model provides instant offline screening and leaf validation. Once connectivity is available, the photo syncs to the backend where a Google Genkit pipeline calls Gemini 3.1 Flash Lite, grounded by Vertex AI Search (RAG) and validated against Pl@ntNet across 80+ crop diseases.

Manager: Reviews the multimodal AI diagnosis and tailored agronomic recommendation, retains full authority to override the classification, and automatically checks recommended chemical dosages against real-time on-farm inventory.
Owner: If warehouse stock is depleted, the system auto-drafts a standardized supplier purchase order and pre-formats a WhatsApp-ready inquiry message. The owner reviews and authorizes the purchase before any order is issued, preserving financial oversight and recording every transaction in an immutable audit log.
Beyond the farm gate, the Farmer Hub maintains offline-synced historical scan records, and a Government Dashboard aggregates regional outbreak trends and commodity price indices.

Responsible AI: AI operates strictly as an assistive diagnostic and drafting co-pilot; it never authorizes funds or replaces certified agronomists.

SDC Workshop Skills Applied: GenAI workflow orchestration (Google Genkit & Gemini), relational database modeling and migrations (MySQL / TiDB Cloud via Drizzle ORM), containerized cloud microservices (Docker on Render, frontend on Vercel), and S3 object storage.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology
Frontend	React 19, TypeScript 5.9, Vite 7, Tailwind CSS 4, Radix UI / shadcn, Recharts, Framer Motion, Wouter
Edge AI	TensorFlow.js 4.22 + MobileNet 2.1.1 (on-device, offline leaf & disease screening)
Backend	Node.js 20+, Express 4.21, tRPC 11.6
Cloud AI	Google Genkit 1.32, Gemini 3.1 Flash Lite, Vertex AI Search (RAG), Pl@ntNet API, Groq vision fallback
Database & Storage	MySQL (TiDB Cloud Serverless) with Drizzle ORM, AWS S3 (image storage), browser IndexedDB (offline sync)
Deployment	Vercel (web frontend), Render (Dockerized backend API)

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: We mapped the operational chain of Malaysian smallholder farms (field worker, supervisor, shed inventory, chemical dealer) and identified recurring friction points in informal WhatsApp reporting and inventory stock-outs.

Define: The primary bottleneck is the 48-hour diagnostic-to-procurement delay caused by scarce on-site pathology expertise and disconnected chemical stock records.

Ideate: We prioritized a focused, role-based tool over complex enterprise ERPs, restricting AI to vision diagnosis and structured drafting while enforcing strict human approval for purchasing decisions.

Prototype: Built a responsive web app with synthetic farm data, on-device Edge AI for offline rural connectivity, automated inventory matching, and a Telegram bot for urgent notifications.

Test: Conducted internal usability benchmarks targeting sub-30-second leaf scans and sub-1-minute purchase order drafting and owner sign-off.

Deploy: The application is fully containerised with Docker on Render and hosted on Vercel, featuring type-safe tRPC APIs, automated fallback models, and role-based access control—ready for immediate live deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s)	Target 2.3: Double the agricultural productivity and incomes of small-scale food producers through reduced crop loss. Target 2.4: Ensure sustainable food production and resilient agricultural practices against climate-driven pests. *(Supporting: Target 8.2 on technological upgrading).*
Current Problem Baseline	(a) Rice blast alone causes historical yield losses of 10%–50% in severe Malaysian epidemics across Perak, Penang, Selangor, and Kedah. (b) The average delay between noticing a field anomaly and ordering chemical treatment is ~48–72 hours on smallholder farms.
Core Measurable KPI	Elapsed time from leaf photo upload to supplier purchase order approval (minutes/hours), and % adherence to recommended chemical dosage rates.
Phase 2 Impact Target	≥95% reduction in detection-to-order turnaround time (from ~48 hours to under 2.4 hours in simulated operations; sub-5 minutes during live demos), with 100% SOP-compliant dosage calculations.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type(code/dataset/model)	License
React, Vite, Tailwind CSS, tRPC, Express, Radix/shadcn, Recharts, Framer Motion, Wouter	Code	MIT / ISC
TensorFlow.js	Code	Apache-2.0
MobileNet v1 (2.1.1)	Model (pre-trained)	Apache-2.0
Google Genkit, Drizzle ORM	Code	Apache-2.0
Gemini 3.1 Flash Lite / Vertex AI Search	Model / Service	Google API Terms
Pl@ntNet API	service	Pl@ntNet API terms
World Bank Open Data	Dataset (macro-indicators)	CC BY 4.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



23/10/2026